Exercise Week 3

Advanced Applied Statistics

Fall 2025

Problem 1*

Let X be an exponentially distributed random variable with the probability density function $f_\lambda(x) = \lambda e^{-\lambda x}, x > 0$. Show that $\int_0^\infty f_\lambda(x)dx = 1$.

Problem 2*

Assume $X_1, \ldots, X_n \sim \text{Bernoulli}(\theta)$. Write down

(a) The likelihood function $L(\theta)$

(b) The log-likelihood function $\ell(\theta)$

(c) The score function (i.e., the derivative of the log-likelihood)

(d) The Fisher information of θ.

Problem 3

In a certain machine learning application, you are working with a large covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{p \times p}$ where p is very large (e.g., $p = 10000$). Instead of inverting the entire matrix, you decide to use the Schur complement for a smaller block of interest.

(a) Explain how you would use the Schur complement to focus on a subset of features, say the first k features.

(b) Provide a practical scenario where this approach would be advantageous.

Problem 4

You have two multivariate normal random vectors $\mathbf{X}$ and $\mathbf{Y}$, whose joint covariance matrix, assuming that the vectors are concatenated as $(\mathbf{X}, \mathbf{Y})$, is

$$\mathbf{\Gamma} = \begin{pmatrix} \mathbf{\Gamma}_{11} & \mathbf{\Gamma}_{12} \\ \mathbf{\Gamma}_{21} & \mathbf{\Gamma}_{22} \end{pmatrix} \tag{1}$$

Assume that you have observed a realisation for $\mathbf{X}$ denoted by $\mathbf{x}$. Write down an expression that characterises the distribution of $\mathbf{Y}$ conditioned on $\mathbf{x}$ that is $\mathbf{Y}|\mathbf{x}$.

Problem 5

Consider a random variable X whose probability distribution belongs to the exponential family. The probability density or mass function of X can be written in the following form:

$$p(x \mid \eta) = h(x) \exp\left(\eta^T T(x) - A(\eta)\right),$$

where:

- x represents the observed data.
- η is the natural parameter (which could be a vector).
- $T(x)$ is the sufficient statistic.
- $A(\eta)$ is the log-partition function (also known as the cumulant function).
- $h(x)$ is the base measure or scaling function.

Write down such a distribution in the exponential family form above by identifying its component, where the distribution is **Exponential distribution** with rate parameter λ:

$$p(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0.$$